

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 12 · Differentiation

Questions

Student book · questions with writing space

- 69 pages
- Past papers up to June 2026

Differentiation

Your classified route

26
questions

208
marks

5
families

10
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Differentiation

Families and variants

F01 • Power-rule differentiation and algebraic preparation

V01 Rewrite then find the first derivative

V02 Second derivative and higher-derivative comparison

F02 • Tangents, normals and linked line geometry

V01 Tangent or normal at a given point

V02 Find the contact point from a gradient or line

V03 Linked normals, tangents and parallel-gradient points

F03 • Equations, inequalities and extrema involving derivatives

V01 Stationary-point equation

V02 Solve a prescribed first- or second-derivative value

V03 Gradient inequality or extremum of the derivative

F04 • Chord gradients and the derivative limit

V01 Chord gradient tending to the tangent gradient

F05 • Global curve synthesis with tangents

V01 Tangent re-intersects the original curve

Power-rule differentiation and algebraic preparation

WHAT THIS FAMILY ASKS

Rewrite radicals and quotients as powers, differentiate term by term and simplify exact results.

VARIANT ROUTE

V01 Rewrite then find the first derivative

V02 Second derivative and higher-derivative comparison

3 Complete Classified questions

17 total marks

1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

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1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \quad x > 0$$

(b) Hence find the exact value of the gradient of the tangent to C at the point where $x = 2$ giving your answer in simplest form.

(Solutions relying on calculator technology are not acceptable.)

(2)

4.

In this question you must show all stages of your working.

(a) Write

$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

in the form

$$y = Ax^p + Bx^q$$

where A , B , p and q are constants to be found.

(4)

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$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

(b) Hence find $\frac{dy}{dx}$ giving each coefficient in simplest form.

(3)

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1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(a) $\frac{dy}{dx}$

(3)

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1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(b) $\frac{d^2y}{dx^2}$

(2)

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Tangents, normals and linked line geometry

WHAT THIS FAMILY ASKS

Use derivative gradients with point-gradient form, reciprocal gradients and linked tangent or normal conditions.

VARIANT ROUTE

V01 Tangent or normal at a given point

V02 Find the contact point from a gradient or line

V03 Linked normals, tangents and parallel-gradient points

11 Complete Classified questions

96 total marks

5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

(a) Find $\frac{dy}{dx}$, giving the answer in simplest form.

(3)

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5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

The point $P\left(4, \frac{11}{3}\right)$ lies on the curve.

(b) Find the equation of the normal to the curve at P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

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1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

(a) Find, in simplest form, $\frac{dy}{dx}$

(3)

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1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

The point P lies on the curve and has x coordinate $\frac{1}{2}$

(b) Find an equation of the normal to the curve at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(5)

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1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(4)

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1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

The point $P(4, 3)$ lies on C .

- (b) Find the equation of the normal to C at the point P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

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1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(3)

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1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

The point $R\left(2, \frac{13}{2}\right)$ lies on C .

(b) Find the equation of the tangent to C at the point R . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

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3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(4)

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3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

The point $P(2, 4)$ lies on C .

(b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

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1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ writing the answer in simplest form.

(3)

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1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

The point $P\left(\frac{1}{4}, \frac{5}{2}\right)$ lies on C .

(b) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$ where m and c are integers.

(3)

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$$y = \frac{1}{2}x^2 - 10x + 22$$
$$a(x + b)^2 + c$$

(3)

$$y = \frac{1}{2}x^2 - 10x + 22$$

The point M is the minimum turning point of C , as shown in Figure 3.

(b) Deduce the coordinates of M

(2)

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Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

The line l is the normal to C at the point P , as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

(c) (i) find the coordinates of P

(ii) find the value of k

(6)

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Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

The line l is the normal to C at the point P , as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

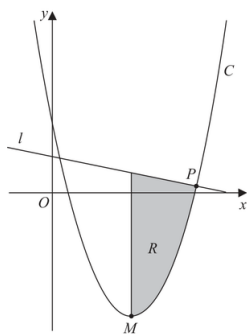


Figure 4

Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by I , C and the line through M parallel to the y -axis.

(d) Identify the inequalities that define R .

(3)

5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The curve C has equation

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(2)

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In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

- the point P lies on C
- the line with equation $y = k - 5x$, where k is a constant, is the tangent to C at P

$$12x^4 + 5x^2 - 2 = 0$$

In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

- the point P lies on C
- the line with equation $y = k - 5x$, where k is a constant, is the tangent to C at P

(c) Hence find the value of k .

(4)

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

(a) Find $\frac{dy}{dx}$

(2)

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$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

(b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)


$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

- (a) Deduce the range of values of x for which $f(x) \geq 0$

(1)

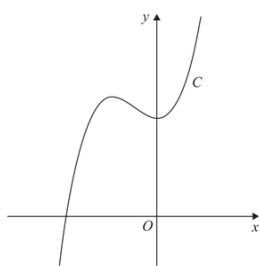


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

(b) Find $f'(x)$ giving your answer in simplest form.

(3)

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$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P .

(4)

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$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

- (d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,
- (i) $y = f(x - 3)$
- (ii) $y = 4f(x)$

(2)

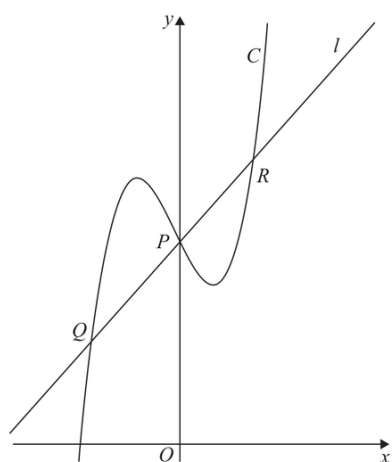


Figure 4

(a) Find, using algebra, the x coordinate of point Q .

(6)

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$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

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$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

Equations, inequalities and extrema involving derivatives

WHAT THIS FAMILY ASKS

Solve conditions imposed on f' or f'' and interpret domain, sign or minimum-gradient information.

VARIANT ROUTE

V01 Stationary-point equation

V02 Solve a prescribed first- or second-derivative value

V03 Gradient inequality or extremum of the derivative

9 Complete Classified questions

73 total marks

9.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$ **(6)**

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3.

In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find $\int f(x) dx$

(4)

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3.

In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(b) (i) Show that when $f'(x) = 0$

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of x for which

$$f'(x) = 0$$

(5)

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7.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(a) Solve the equation

$$f(x) = 9$$

(4)

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7.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(b) Solve the equation

$$f''(x) = 6$$

(5)

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2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = 3x^5 + 4x^3 - x + 5$$

The points P and Q lie on the curve.

The gradient of the curve at both point P and point Q is 2

Find the x coordinates of P and Q .

(5)

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7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

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7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

(b) find the value of k .

(2)

(c) Hence find the coordinates of B .

(3)

2. The curve C has equation

$$y = 2x^{\frac{5}{2}} - 4x + 3$$

(a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(2)

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2. The curve C has equation

$$y = 2x^{\frac{5}{2}} - 4x + 3$$

The point P lies on C .

Given that

- the x coordinate of P is 2^k where k is a constant
- the gradient of C at the point P is 16

(b) find the value of k .

(3)

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4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^3 - 2x^2\sqrt{x} + 14 \quad x > 0$$

(a) Solve the equation

$$f'(x) = 0$$

(5)

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In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

$$x > 0$$

(b) (i) Show that the equation

$$f''(x) = 66$$

can be written as

$$4x - 5x^{\frac{1}{2}} - 44 = 0$$

(ii) Hence, using algebra, show that $x = 16$ is the only solution of the equation

$$f''(x) = 66$$

(5)

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

(a) find the value of the constant p .

(2)

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$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x + q)$ passes through the origin,

(b) write down the possible values of the constant q .

(2)

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$$f(x) = (x + 4)(x - 2)(2x - 9)$$

(c) Find $f'(x)$.

(4)

(d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18

(3)

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x+q)^2+r$$

where p , q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{dy}{dx}$

(ii) the value of x at which this minimum value occurs.

(2)

The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

Given that S is the point on C at which the gradient is a minimum,

- (c) find the equation of the tangent to C at S , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

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Chord gradients and the derivative limit

WHAT THIS FAMILY ASKS

Calculate a chord gradient using an increment h and show that it approaches the tangent gradient as h tends to zero.

VARIANT ROUTE

V01 Chord gradient tending to the tangent gradient

2 Complete Classified questions

12 total marks



The point $P(2,13)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P .

(2)



The point $P(2,13)$ lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form. (3)

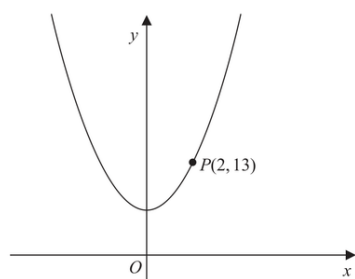


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

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Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

- (2)

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

The point Q with x coordinate $3 + h$ also lies on the curve.

- (b) Find, in terms of h , the gradient of the line PQ . Write your answer in simplest form. (3)

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Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

The point Q with x coordinate $3 + h$ also lies on the curve.

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

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Global curve synthesis with tangents

WHAT THIS FAMILY ASKS

Combine factorised or expanded curve structure with a tangent equation to locate another intersection or interpret the whole curve.

VARIANT ROUTE

V01 Tangent re-intersects the original curve

1 Complete Classified questions

10 total marks


$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

- $$f(x) > 0$$

(2)

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

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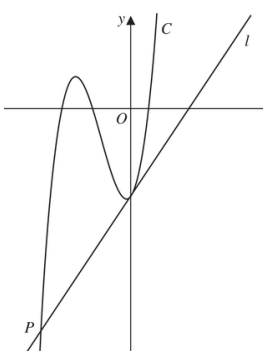


Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

The straight line l is the tangent to C at the point where C cuts the y -axis.

Given that l cuts C at the point P , as shown in Figure 4,

(c) find, using algebra, the x coordinate of P

(Solutions based on calculator technology are not acceptable.)

(5)

